

Anomaly and Fault Classification and Prediction using Photovoltaic Digital Twins and Machine Learning based approaches

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This paper presents a hybrid framework that integrates photovoltaic (PV) digital twins with machine learning (ML) techniques for anomaly detection, fault classification, and predictive maintenance in PV systems. Owing to the scarcity of labelled real-world fault data, digital twins are developed using the PVlib Python library to simulate seven operational states: normal operation, three anomalies (soiling, shading, and cracks), and three electrical faults (ground fault, arc fault, and bypass diode failure). Synthetic daily time-series are generated to train and evaluate multiple classifiers, with XGBoost achieving superior performance, reaching 82.9% overall accuracy on the synthetic test set and up to 100% accuracy for distinct electrical faults. A probabilistic inference framework is introduced to quantify prediction uncertainty and support informed maintenance decision-making. The proposed model is further validated using real operational data, showing that the resulting fault and anomaly distributions align with realistic operating scenarios and are representative of conditions observed in real-world PV systems. For predictive analysis, a linear regression-based forecasting model estimates anomaly progression from daily class probabilities, achieving success close to 100% and an anticipation horizon of approximately 30 days under Pareto-optimal parameter configurations identified through sensitivity analysis. Overall, the results demonstrate the effectiveness and practical applicability of the proposed approach for enhancing reliability and predictive maintenance capabilities in PV systems, even in scenarios with limited real fault data.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Supervised learning**; *Anomaly detection*; • **Applied computing** → *Digital twins*; • **Information systems** → Data mining; • **Hardware** → Energy distribution and storage; • **Software and its engineering** → Software maintenance.

Additional Key Words and Phrases: Photovoltaic Systems, Digital Twins, Predictive Maintenance, Machine Learning, Anomaly and Fault Classification, Time-Series Prediction

ACM Reference Format:

Ricardo Vicente. 2026. Anomaly and Fault Classification and Prediction using Photovoltaic Digital Twins and Machine Learning based approaches. 1, 1 (February 2026), 10 pages.

1 Introduction

The global energy landscape is shifting as renewable adoption grows to cut emissions and enhance energy independence [Atstāja 2025]. Photovoltaic (PV) systems are central to this transition due to sustainability and falling costs [Yang et al. 2019], yet long-term reliability is challenged by anomalies and faults. Maintenance is moving from preventive schedules to predictive approaches using data analytics and machine learning (ML) [Huuhtanen and Jung 2018], but the scarcity of labelled fault data limits real-world applicability [Thakfan and Salamah 2024].

To address this, we propose a hybrid digital twin and ML framework for PV fault classification and prediction. Using the open-source PVlib Python, physically-grounded digital twins generate realistic synthetic data for short-term anomaly and fault classification, and long-term degradation forecasting. Seven operational states are simulated, including normal operation, three anomalies (soiling, shading, cracks), and three electrical faults (ground, arc, bypass diode). The two-stage pipeline integrates state classification and time-series prediction of fault progression, with probabilistic inference for diagnostics, uncertainty quantification, and maintenance decision support.

1.1 Background

PV systems convert sunlight into electricity and consist of PV modules, inverters, and optional energy storage [Chowdhury et al. 2025]. The inverter converts DC to AC and implements Maximum Power Point Tracking (MPPT) to maximise energy yield under varying conditions [Iturralde Carrera et al. 2025]. Reliable operation requires monitoring electrical and environmental variables, as PV systems face stressors that degrade performance or cause failure. These are classified as anomalies—deviations without immediate shutdown, such as efficiency losses from soiling—and faults, which are physical or logical defects potentially causing partial or total failure [Mariani et al. 2024].

Common anomalies include soiling and shading. Soiling from dust or contaminants can reduce power output by up to 50%, primarily affecting current-related parameters [Abid et al. 2018]. Shading from obstacles or clouds induces electrical mismatches, activates bypass diodes, and may create hotspots, accelerating degradation [Abdelaziz et al. 2022]. Cracks and microcracks in silicon cells, caused by mechanical or thermal stress, progressively reduce effective cell area and power output [Makhija and Bohra 2023].

Electrical faults are more severe and often safety-critical. Ground faults create leakage paths increasing fire risk [Brooks 2015], arc faults arise from insulation breakdown or loose connections with high-frequency current oscillations [Gu et al. 2019], and bypass diode faults alter voltage characteristics, potentially causing localised overheating [Mühleisen et al. 2024].

1.2 Related Work

Despite significant methodological advances in PV system monitoring, substantial gaps remain between academic research and practical deployment. A central issue concerns the trade-off between data availability and model generalisation. Most studies rely on supervised learning, achieving high accuracy under controlled conditions but requiring labelled datasets that are rarely available in real PV installations [Livera et al. 2022]. Although deep learning (DL) models such as convolutional and long short-term memory networks often report accuracies above 97% on synthetic data [Appiah

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et al. 2019], their performance typically degrades in operational environments due to environmental variability, sensor noise, and the lack of representative labelled faults [Sepúlveda-Oviedo 2023]. This dependence on simulated or limited real data creates a pronounced reality gap between laboratory validation and field deployment.

Unsupervised and semi-supervised approaches partially address label scarcity, but introduce new limitations. Methods such as autoencoders and isolation forests can detect deviations from normal operation [Hempelmann et al. 2020], yet they generally fail to provide meaningful fault identification, offering limited diagnostic insight for maintenance decision-making [Bakdi et al. 2020]. Their sensitivity to environmental fluctuations often results in elevated false alarm rates, reducing reliability in practical settings [Kieu et al. 2022]. Hybrid frameworks attempt to improve diagnostic capability by combining learning paradigms, but frequently suffer from high computational complexity and extensive parameter tuning, hindering deployment in real-time and resource-constrained PV systems [Zhao et al. 2015].

A critical and underexplored gap lies in predictive capability. While fault detection and classification are widely studied, few works address the temporal forecasting of fault progression [Ledmaoui et al. 2025]. Existing predictive models often rely on simplified regression or basic time-series methods that inadequately capture the non-linear degradation behaviour of PV anomalies [Brahma et al. 2022]. Moreover, uncertainty quantification and interpretability—key requirements for maintenance planning—are largely absent from current approaches [Fonoberova et al. 2013].

Additional open challenges include the handling of concurrent and compound faults, which are common in real PV systems but seldom modelled [Liu et al. 2025a], and the computational burden of DL methods in edge or inverter-level monitoring contexts [Buratti et al. 2024; Karthikeyan and Jagadeeshwaran 2023]. Finally, the lack of standardised, publicly available benchmark datasets with comprehensive fault annotations continues to impede fair comparison and methodological progress [Rodrigues 2024]. These gaps collectively motivate the hybrid digital twin and ML framework proposed in this work.

1.3 Main Contributions and Paper Organization

This dissertation makes several key contributions to the field of predictive maintenance for PV systems. The primary contribution is a novel hybrid framework that integrates PV digital twins with ML to tackle the critical challenge of scarce real-world fault data. The framework is realized through two main pipelines: one for classification and another for prediction. Key specific contributions include:

- (1) **Digital Twin-Based Data Generation:** PV digital twins are developed using PVlib to generate labelled synthetic data for normal operation, anomalies, and electrical faults, enabling model training and validation.
- (2) **Robust Anomaly and Fault Classification:** Multiple classifiers are evaluated on synthetic data, with XGBoost achieving the best performance, particularly for electrical faults, and enabling feature importance analysis.

- (3) **Probabilistic Inference Framework:** A probabilistic inference approach provides interpretable predictions with uncertainty quantification and is validated on real operational data.
- (4) **Predictive Capability for Anomaly Progression:** The framework is extended to predict anomaly evolution using linear regression on class probabilities, with Pareto-optimal configurations identified via sensitivity analysis.

Starting from the foundational context, this paper first established the necessary background on PV systems and reviewed related work, highlighting existing gaps in fault detection and prediction. It then outlines the main contributions before introducing the proposed hybrid digital twins and ML framework. The subsequent sections detail the framework's implementation—including the digital twin setup and data generation—followed by the experimental methodology and evaluation of classification and prediction results. The paper concludes with a discussion interpreting these results, acknowledging limitations, and suggesting directions for future research.

The source code resulting from this dissertation, including the developed digital twin models and the classification and prediction pipelines, is publicly available on GitHub at PV Digital Twins for Classification and Prediction.

2 Hybrid Framework for Predictive Maintenance

The proposed hybrid framework integrates PV digital twins with ML to form an end-to-end pipeline for anomaly and fault classification and prediction in PV systems. As shown in Figure 1, the architecture comprises two interconnected computational streams driven by distinct digital twin simulations. The first stream addresses classification, using daily synthetic data to train and evaluate ML models across seven operational states: normal operation, three anomalies (soiling, shading, cracks), and three electrical faults (ground, arc, bypass diode). The second stream focuses on prediction, applying the best-performing classifier to time-series data to forecast the evolution of degradative conditions via linear regression.

2.1 Digital Twins

Digital twins serve as a foundational component of the framework, providing labelled synthetic data for training ML models when real-world fault examples are scarce. A digital twin is a virtual representation of a physical system that enables monitoring, simulation, and predictive analysis through continuous data integration [Jain et al. 2019]. In PV systems, digital twins enhance management and maintenance by enabling controlled simulation of fault conditions and performance scenarios. When combined with ML, this technology has demonstrated substantial benefits across the PV lifecycle, including performance optimisation, predictive maintenance, and fault diagnosis, with reported reductions of 35% in unplanned downtime, an 8.5% increase in energy production, and fault detection accuracy of 98.3% [Abdessadak et al. 2025]. Digital twins also support energy management and forecasting by simulating usage scenarios and optimising resource allocation in microgrid environments [Angelova et al. 2024].

Despite their potential, widespread adoption of digital twins faces challenges including high costs for sensors and computing

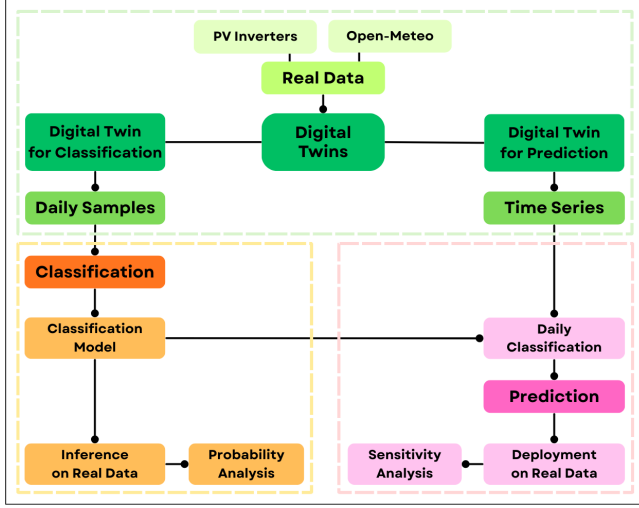


Fig. 1. Diagram of the hybrid classification and prediction model supported by digital twins.

infrastructure, integration complexity with legacy systems, and cybersecurity risks [Abdessadak et al. 2025]. Other limitations include heterogeneous data management, environmental impacts from high computational energy use, and modelling inaccuracies that may overlook effects such as soiling, degradation, or shading [Guzman Razo et al. 2020]. To ensure practical feasibility, this work uses the PVlib Python library as the core simulation engine. PVlib is an open-source, widely validated tool for simulating PV electrical performance based on irradiance, temperature, and system parameters. Its capacity to parameterise PV systems accurately, even with limited prior knowledge via methods like Genetic Algorithm optimisation, supports digital twin development. The framework’s digital twins simulate short-term anomalies and faults for classification and long-term degradation for prediction, producing realistic synthetic datasets that capture diverse failure signatures while accounting for environmental variability.

The framework implements two digital twins with distinct purposes. The first generates daily time series for classification, producing balanced datasets for seven operational states across full calendar years. The second produces continuous time-series data over several months to model the progressive evolution of anomalies such as soiling, shading, and cracks. Both digital twins adopt identical system specifications—Huawei SUN2000-10KTL inverters and Jinko JKM410M modules—thereby ensuring consistency with the real-world data on which inference is conducted. The simulation accounts for temporal characteristics of anomalies and faults, recognising that some failures appear abruptly while others develop gradually, affecting detection windows. By generating daily samples rather than full time-series for classification, the framework reduces computational cost while still extracting statistical features from voltage, current, power, and environmental variables, providing sufficient discriminative information for accurate classification despite lower temporal resolution.

2.2 Classification

The classification pipeline transforms synthetic daily data from the first digital twin into a trained model for identifying PV system states. Traditional ML models are prioritised over DL for faster training, lower computational demands, and superior explainability, all critical for industrial deployment. Four algorithms were evaluated: Naïve Bayes, Decision Trees, Random Forests, and XGBoost. Naïve Bayes provides a probabilistic baseline assuming feature independence. Decision Trees are interpretable but prone to overfitting, while Random Forests improve generalisation via ensemble learning on random data subsets [Breiman 2001]. XGBoost refines this through sequential boosting, regularisation, and efficient computation, achieving high accuracy in predictive modelling [Chen and Guestrin 2016].

Evaluation on the 7-class balanced dataset (2,555 samples) used accuracy, precision, recall, and F1 scores. The best model was then applied to real operational data to assess generalisation, identifying anomalies and faults from Huawei’s Smart platform, which provides a generic fault label (`inverter_state=768`) without specifying type. Probabilistic outputs enable continuous confidence assessment across all classes, allowing maintenance personnel to make informed decisions even under ambiguous conditions, addressing the need for interpretable diagnostics in operational environments.

2.3 Prediction

The prediction pipeline extends the classification framework to forecast anomaly progression over time, using the best-trained classifier to analyse time-series data from the second digital twin. The approach involves three stages: daily classification using the model from the training phase; and forecasting class evolution via linear regression, employing a sensitivity analysis of the regression parameters to optimize reliability. Initially applied to synthetic anomaly series, it was then implemented on smoothed real data from Huawei’s Smart platform. Smoothing mitigated cloud-induced fluctuations, preserving sensitivity to genuine degradation trends and supporting preventive maintenance planning.

The first stage loads the pre-trained XGBoost classifier and applies it to daily time-series data. Model outputs are probability distributions across seven classes, forming the basis for trend analysis. The second stage predicts when a non-normal class probability reaches a defined threshold. `LinearRegression` fits probabilities from the previous window days, extrapolating the date the `threshold_target` will be reached, considering only positive slopes to reflect worsening conditions.

In this stage, a sensitivity analysis is performed on four key parameters: `threshold_start`, `threshold_target`, `threshold_class`, and `window` [Pierro et al. 2022]. Grid search evaluates combinations, measuring prediction success rate and mean lead time. Pareto charts identify optimal configurations balancing early detection and accuracy [Giannelos et al. 2024], enabling customization for gradual anomalies like soiling or shading. The framework thus offers an efficient, deployable approach to predictive maintenance in resource-constrained PV monitoring environments.

3 Digital Twins Implementation

Digital twins for PV system simulation were implemented using the PVlib Python library to create realistic synthetic datasets for classification and prediction. The process translated real-world system specifications and operational data into computational models reproducing electrical behaviour under normal and abnormal conditions. This involved integrating meteorological measurements, replicating the PV system layout, and systematically introducing anomaly and fault signatures. The resulting digital twins produced labelled datasets capturing seven operational states, enabling ML model development and evaluation.

3.1 Real Data

The implementation used two complementary real-world datasets: environmental data from the Open-Meteo API and operational data from Huawei's Smart PV platform. These datasets served three key purposes: meteorological variables (GTI, DHI, temperature, wind speed) drove the simulation engine [Koohestani et al. 2023]; environmental and electrical measurements formed the feature set for model inference on real states; and synthetic outputs, particularly power parameters, were validated against actual measurements to confirm the digital twin's fidelity before introducing faults.

The retrieved Huawei dataset included five-minute interval data from 80 string inverters (May 9–Aug 26, 2025), covering voltage (`pv1_u`), current (`pv1_i`), DC power (`mppt_power`), AC power (`active_power`), inverter temperature, and operational state (`inverter_state`). Data with the label 768, which indicated an unspecified fault, was extracted for classification. Missing values were addressed via hierarchical imputation: major gaps replaced by previous-day values, moderate gaps by linear interpolation or forward filling; negative values were set to zero. A rolling window (`window=48`, `min_periods=1`) smoothed cloud-induced fluctuations. Correlation analysis confirmed expected relationships: PV1 current correlated with GTI ($r=0.66$) and DC power ($r=0.59$), while inverter temperature correlated moderately with irradiance ($r=0.55$) and current ($r=0.51$).

3.2 Digital Twin Layout

The digital twin layout was constructed to mirror a physical PV installation located in Northern Portugal, utilizing specifications for the Huawei SUN2000-10KTL-USL0-240V inverter and Jinko JKM410M-72HL-V solar modules. This location was selected for its climatic variability, which introduced greater parameter diversity into the simulated datasets. The system configuration was implemented through several computational classes within the PVlib framework. The `pvlib.location` class, which set latitude, longitude, and timezone for solar position calculations. The `PvPlant` class encapsulated the physical array layout, specifying the number of modules per string, strings per inverter, surface tilt, azimuth orientation, and albedo coefficient, creating a `PVSystem` object that represented the physical array within the simulation environment. The `InverterTwin` class modelled the power conversion behaviour using the Sandia inverter model, simulating DC to AC conversion dynamics.

The core simulation logic was implemented in a `DigitalTwin` superclass, which combined meteorological data with system specifications to estimate energy generation. The algorithm began by initializing a `ModelChain` object for PV simulation, then computed solar position and irradiance components. The `Erbs` function, implementing the ERBS solar radiation decomposition model [Erbs et al. 1982], estimated direct normal irradiance (DNI) and DHI from measured GTI. The `ModelChain` executed the simulation, calculating DC and AC power, system efficiency, and thermal conditions. Outputs included DC power, AC power, efficiency, phase voltages and currents, and inverter temperature, providing a comprehensive representation of system electrical behaviour. For normal operation simulations, meteorological data for 2023 (training set) and 2024 (test set) were processed in five-minute intervals, with daily extensive numerical features extracted according to the feature engineering dictionary. These features included basic statistics (e. g., max, mean, median, std), temporal differences (e. g., positive/negative counts, mean differences), and rolling window descriptors (e. g., mean, volatility, correlation), capturing a wide range of data characteristics for ML model training.

3.3 Digital Twin for Classification

For classification tasks, digital twins were developed to simulate seven distinct operational states: normal condition, three anomalies (soiling, shading, cracks), and three electrical faults (ground fault, arc fault, bypass diode fault). Each condition was represented by a dedicated subclass extending the `DigitalTwin` superclass, with specific parameter modifications introduced to replicate the physical effects of each anomaly or fault type. For each condition, 365 daily simulations were generated for 2023 (training) and another 365 for 2024 (testing), resulting in balanced datasets of 2,555 samples each.

Soiling was simulated through the `DigitalTwinWithSoiling` subclass, which introduced a daily soiling loss factor between 5% and 30% to represent dust accumulation. This factor primarily reduced current and power outputs while leaving voltage largely unaffected, consistent with the physical behaviour of soiling [Arani and Akhavanhejazi 2016]. Precipitation events exceeding 2.1 mm triggered partial cleaning, reducing accumulated soiling by 40% to simulate natural cleaning effects, creating dynamic daily profiles that reflected both accumulation and intermittent cleaning.

`DigitalTwinWithShading` subclass was used to implement shading conditions, applying a daily shading factor between 0.05 and 1.0 to represent partial obstruction of sunlight. This factor reduced both current and voltage, with a typical voltage reduction of 20% under partial shading conditions, reproducing the electrical mismatch effects characteristic of shaded PV strings [Teo et al. 2018].

`DigitalTwinWithCracks` subclass was employed to simulate crack degradation, utilizing random degradation factors to current (0.910–0.990), voltage (0.930–0.985), and power (0.750–0.970), representing performance reductions due to mechanical fractures in silicon cells [Bdour et al. 2020]. These factors were applied uniformly throughout each simulated day, producing consistent degradation across all electrical parameters.

Electrical faults required more complex simulation approaches due to their abrupt nature and safety implications. Ground faults

were simulated using the `DigitalTwinWithGround` subclass, which introduced both gradual degradation preceding the fault and an abrupt system shutdown at a randomly selected daily time. Prior to shutdown, current and voltage were degraded according to randomly selected factors (current: 0.30–1.02; voltage: 0.80–0.97), representing progressive insulation weakening and leakage effects [Flicker and Johnson 2016]. At the fault moment, all electrical parameters dropped to zero to simulate protection system intervention.

Arc faults were implemented via the `DigitalTwinWithArc` subclass, which introduced sudden voltage increases (1–5 V) and current reductions (30–50%) during a one-hour fault window, followed by complete system shutdown. This approach replicated the high-impedance characteristics of electrical arcs while acknowledging the limited temporal resolution (5-minute intervals) available for capturing high-frequency arc signatures [Lu et al. 2018].

`DigitalTwinWithDiode` subclass was employed to simulate bypass diode faults, reducing DC voltage according to the number of bypassed substrings while maintaining near-constant current, reproducing the electrical behaviour of malfunctioning bypass diodes [Xiao et al. 2020].

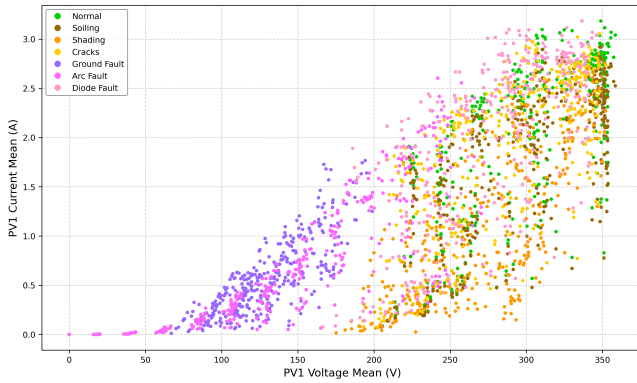


Fig. 2. Scatter I-V Plot of the Training Set (2023).

Figure 2 presents the scatter I-V visualization of the consolidated training dataset, which reveals overlapping clusters for normal and anomalous operating conditions but clearer separation for electrical faults, particularly ground and arc faults, highlighting both the challenges and opportunities for ML classification.

3.4 Digital Twin for Prediction

For prediction tasks, a digital twin was developed to simulate long-term degradation trends over six-month periods (April to September), focusing exclusively on the three anomaly types as electrical faults were considered impractical for long-term forecasting from electrical data alone. Three dedicated subclasses were created, each implementing time-dependent degradation mechanisms that evolved throughout the simulation period.

The `DigitalTwinSoilingTS` subclass simulated soiling accumulation through a growth rate that increased the soiling factor up to a predefined maximum. Precipitation events exceeding a threshold

triggered partial cleaning (`cleaning_efficiency`), reducing accumulated soiling and creating the irregular progression patterns characteristic of real-world soiling accumulation influenced by weather conditions [Brahma et al. 2022]. The soiling factor constrained between minimum and maximum limits was applied to reduce current and power outputs, producing realistic energy loss profiles that reflected both gradual accumulation and intermittent cleaning.

The `DigitalTwinShadingTS` subclass simulated progressive shading effects through an initial shading fraction that increased gradually according to a fixed growth rate. At each timestamp, this fraction reduced module current and, to a lesser extent, voltage, representing the expanding obstruction of sunlight over time. These adjustments produced continuous degradation profiles that captured the slow progression of shading conditions, whether from vegetation growth or structural changes in the surrounding environment.

The `DigitalTwinCracksTS` subclass implemented crack progression through degradation scenarios specifying final reduction factors for voltage, current, and power. These factors evolved according to a normalized exponential curve, starting with minor initial reductions and accelerating toward the final degradation values as cracks propagated. This model simulated the physical behaviour of mechanical fractures, where early-stage damage causes minimal performance impact that intensifies as cracks expand and interconnect [Papargyri et al. 2022].

All three subclasses generated both affected and clean system outputs for comparison, enabling direct quantification of performance losses over time. The resulting time-series datasets captured the temporal dynamics of anomaly progression under varying environmental conditions, providing the foundation for developing and evaluating predictive maintenance algorithms that could forecast degradation trajectories based on historical patterns.

4 Experiments Evaluation

The experimental evaluation was structured to validate the proposed hybrid framework across both classification and prediction tasks. A systematic methodology was followed, beginning with the comparison of multiple ML models on synthetic data to identify the optimal classifier. This was followed by statistical validation of model performance, inference on real-world operational data, and finally, the application of the selected model to time-series forecasting for anomaly progression. The evaluation employed a range of metrics including accuracy, precision-recall analysis, confidence intervals, and sensitivity analysis to provide a comprehensive assessment of the framework’s capabilities and limitations under realistic conditions.

4.1 Experimental Setup

All models were trained on a system with an AMD Ryzen 7 7435HS processor (16 cores), 24GB RAM, and an NVIDIA GeForce RTX 4070 GPU, running Windows 11. This setup supported efficient multi-threading and parallel processing, ideal for large datasets.

4.2 Classification

The classification evaluation commenced with training and testing multiple ML algorithms—Naïve Bayes, Decision Trees, Random

Forests, and XGBoost—on synthetic daily data generated by the classification digital twin. The 2023 dataset was split into training (70%) and validation (30%) sets with stratification to preserve class distributions, while the 2024 dataset served as an independent test set. Initial performance metrics revealed that Random Forests and XGBoost achieved the highest scores across precision, recall, and F1 measures on both validation and test sets (Figure 3). Given their similar performance, a statistical comparison was conducted to determine whether the observed difference in accuracy was significant. A 5-fold cross-validation was performed on both models using identical folds. A paired t-test was applied to the cross-validation results, yielding a t-statistic of approximately 3.867 and a p-value of 0.018, which was below the 0.05 significance threshold. The Shapiro-Wilk test for normality indicated that the differences were approximately normally distributed ($W=0.8857$, $p=0.336$), supporting the use of the parametric test despite the limited sample size. These results led to the selection of XGBoost as the superior model for all subsequent pipeline operations.

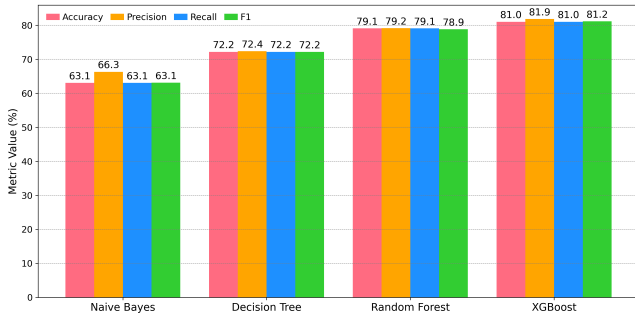


Fig. 3. Comparison of Models (Test Set).

The XGBoost classifier was configured with specific hyperparameters: $n_estimators=200$, $max_depth=6$, $learning_rate=0.1$, $subsample=0.8$, $colsample_bytree=0.8$, and $random_state=42$. The objective was set to multi-class probability estimation (multi:softprob) with logarithmic loss as the evaluation metric. The trained model achieved an overall accuracy of 80.5% on the validation set and 82.9% on the test set, substantially higher than the 14.28% expected from random guessing across seven balanced classes. Analysis of the confusion matrices (e.g., Figure 4) revealed that electrical faults—particularly ground and arc faults—were classified with near-perfect accuracy due to their distinctive electrical signatures. In contrast, anomalies such as soiling, shading, and cracks exhibited greater inter-class confusion, with soiling often misclassified as normal operation, and cracks occasionally confused with bypass diode faults. This pattern was reflected in the per-class accuracy plots (e.g., Figure 5), which showed electrical faults achieving consistently high scores while anomalies demonstrated more variable performance.

A stratified bootstrap procedure with 1,000 iterations was employed to estimate 95% confidence intervals for each class's accuracy. This analysis revealed narrow confidence intervals for ground faults (3.1%) and arc faults (1.4%), indicating high prediction stability, while wider intervals were observed for soiling (6.5%) and cracks (6.6%), reflecting greater classification uncertainty for these anomaly types.



Fig. 4. Confusion Matrix of the Test Set.

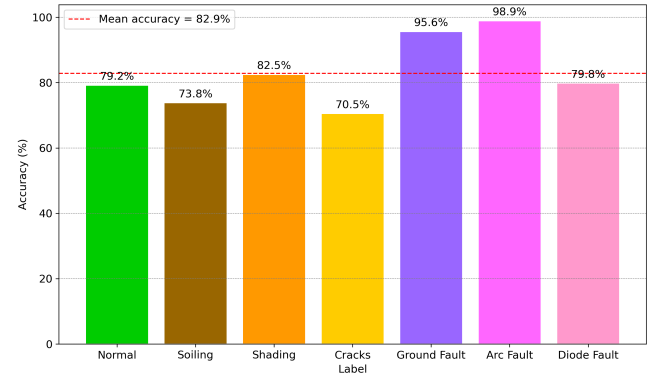


Fig. 5. Per-Class Accuracy of the Test Set.

Feature importance analysis identified electrical variables—particularly string voltage ($pv1_u$)—as the most influential predictors, with meteorological features such as GTI, DHI, and precipitation appearing lower in the ranking but remaining relevant for environmental anomaly detection.

Inference was subsequently performed on real operational data labelled with the generic fault indicator $inverter_state=768$ (Figure 6). The trained model classified approximately 73% of real samples as normal, with the remaining distributed primarily across soiling (12%), shading (1%), and bypass diode faults (15%)—a distribution consistent with typical PV system behaviour where electrical faults are rare and environmental anomalies more frequent [Dhimish et al. 2017].

Probability distributions for individual days demonstrated varying degrees of classification confidence, with some days showing clear dominant classes (e.g., bypass diode fault at 91.9%, in Figure 7)

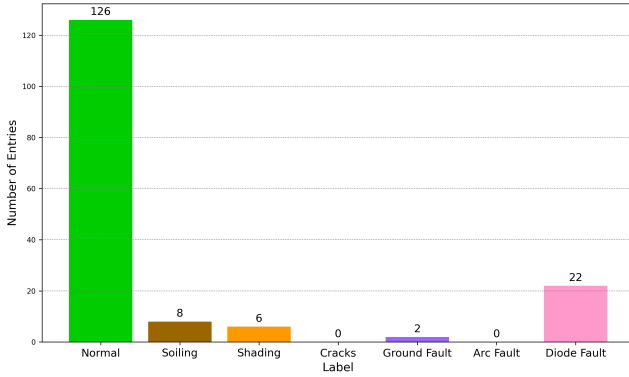


Fig. 6. Class Distribution after Inference on Real Data.

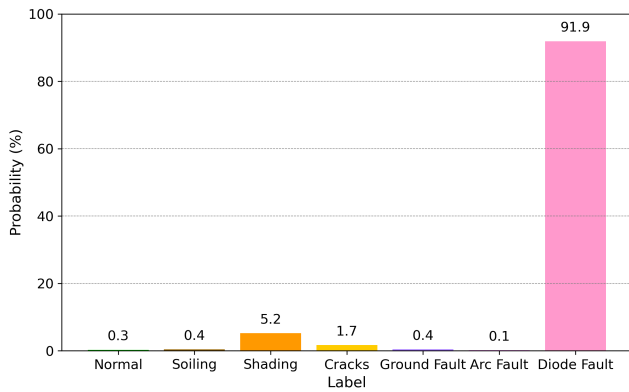


Fig. 7. Probability Distribution Per-Class (Central Portugal, 2025-07-26).

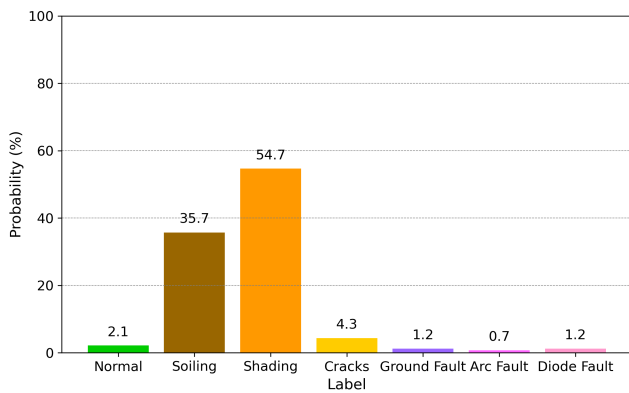


Fig. 8. Probability Distribution Per-Class (Central-Northern Portugal, 2025-05-09).

while others exhibited more ambiguous probability spreads (Figure 8). A probabilistic adjustment procedure was applied to enhance interpretability for maintenance personnel, selectively rescaling probabilities for the top N (e.g., 2) classes within a specified delta

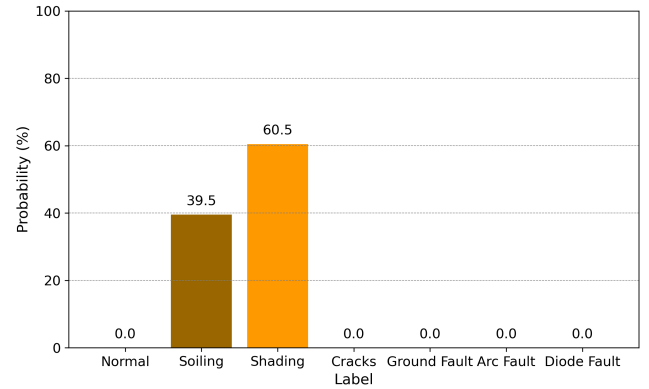


Fig. 9. Adjusted Probability Distribution Per-Class (Central-Northern Portugal, 2025-06-08).

threshold (e.g., 0.2) to produce cleaner, more actionable outputs (Figure 9).

4.3 Prediction

The prediction evaluation applied the trained XGBoost classifier to time-series data from both synthetic and real sources. For synthetic anomaly progression, daily classification probabilities were computed for soiling accumulation, shading worsening, and crack development scenarios. Linear regression was then applied within rolling time windows to forecast when a specific anomaly's probability would exceed predefined thresholds. A sensitivity analysis was conducted through discrete values in a grid search across four parameters: threshold_start (0.2–0.5), threshold_target (0.5–0.8), threshold_class (0.1–0.3), and window (30–80 days). Pareto charts were generated for each anomaly type to identify optimal configurations that balanced early detection (mean lead time) with prediction reliability (success rate).

For soiling accumulation, the Pareto-optimal front included configurations achieving between 76.6% and 100% success rates, with mean lead times ranging from 15 to 30 days, respectively. A configuration emphasising accuracy (threshold_start=0.3, threshold_target=0.5, threshold_class=0.2, window=60) achieved 100% success but with later detection, while a configuration prioritising early warning (threshold_start=0.5, threshold_target=0.8, threshold_class=0.3, window=45) provided 30 days of lead time at 76.6% success. Shading worsening predictions achieved similar performance, with optimal configurations yielding success rates up to 100% and lead times up to 27.7 days. Crack development forecasting proved slightly more challenging due to the non-linear nature of crack propagation, yet still achieved 90% success under optimal parameters (threshold_start=0.4, threshold_target=0.8, threshold_class=0.3, window=30).

The prediction framework was subsequently applied to two real time-series from May to August 2025. Pareto analysis for these real cases identified optimal parameter sets that achieved 100% prediction accuracy with lead times of 6 and 29 days for soiling and

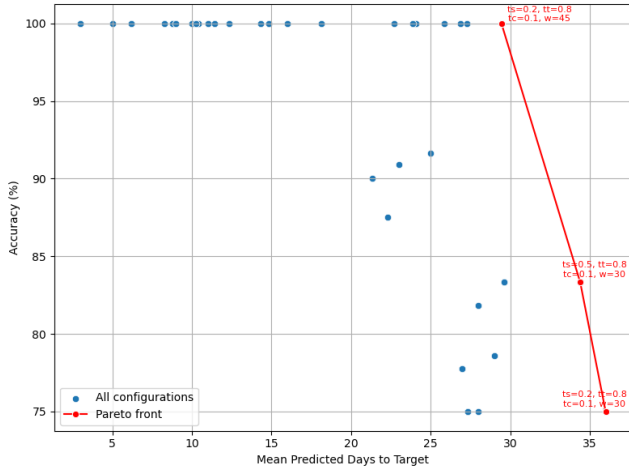


Fig. 10. Pareto Chart for Grid Simulations with Real Data (Central Portugal, May–Aug 2025).

bypass diode faults, respectively. For the latter case, Figures 10 and 11 present the Pareto chart and the corresponding forecasts obtained with an optimal parameter configuration. Although electrical faults were not explicitly modelled in the prediction digital twin, the successful application to real data suggested the framework’s potential for forecasting gradual equipment degradation resulting from fault accumulation.

4.4 Discussion

The experimental results demonstrated that the hybrid digital twin and ML framework effectively addressed key challenges in PV system monitoring. The classification component achieved robust performance despite being trained exclusively on synthetic data, with XGBoost significantly outperforming alternative algorithms. Strong performance on electrical faults aligned with previous studies showing that distinct electrical signatures enable reliable detection [Appiah et al. 2019; Gao and Wai 2020], while the greater difficulty in distinguishing anomalies reflected their overlapping effects on system parameters [Chang et al. 2024; Zhao et al. 2019]. Feature importance analysis confirmed that voltage-based features are particularly discriminative for fault diagnosis, as they are less sensitive to environmental variability than current or power measurements [Dang et al. 2024].

Real-world inference produced plausible class distributions consistent with expected operational profiles, with normal operation dominating and diode bypass fault emerging as the most frequent abnormality. Probabilistic adjustment of classification outputs improved interpretability for maintenance decisions, addressing the need for uncertainty-aware diagnostics in industrial contexts [Fonoberova et al. 2013; Livera et al. 2022]. The successful transfer from synthetic training to real inference supported the digital twin’s ability to generate representative data, although the lack of verified fault labels in the real dataset limited definitive validation of diagnostic accuracy.

The prediction framework extended the classifier’s utility by enabling forward-looking forecasts of anomaly progression. Sensitivity analysis across parameter configurations yielded prediction success rates of up to 100% for selected Pareto-optimal settings, with mean lead times of up to 30 days depending on anomaly type and parameters. These results showed that even linear regression could capture degradation trends when driven by well-calibrated daily probabilities, provided suitable thresholds and window sizes were applied. The analysis highlighted trade-offs between early detection and false alarms, with parameter choices directly affecting operational usefulness. These findings were consistent with sensitivity studies in related predictive maintenance domains [Burg Demay et al. 2024; Pierro et al. 2022] and underscored the importance of context-aware configuration.

Key limitations included the linear regression assumption of monotonic trends, which may not capture non-linear degradation or abrupt fault events. The 5-minute temporal resolution also restricted detection of high-frequency phenomena critical for some electrical faults. Additionally, the focus on isolated anomalies rather than compound fault scenarios limited realism, as real installations often experience concurrent issues with interacting signatures [Liu et al. 2025b]. Nevertheless, the results confirmed the framework’s viability as a practical predictive maintenance solution for PV systems, balancing analytical capability with computational efficiency suitable for operational deployment.

5 Conclusions, Limitations and Future Work

This section summarises the core outcomes, current constraints, and future directions of the hybrid digital twin and ML framework for PV predictive maintenance. It confirms the approach’s validity in tackling data scarcity through synthetic simulation, acknowledges the existing gaps between simulation and real-world deployment, and outlines targeted research to bridge these gaps for practical application.

5.1 Conclusions

This research demonstrated that a hybrid framework combining physically-grounded digital twin simulations with robust ML models offered an effective approach for predictive maintenance in PV systems. A core goal was using the PVlib Python library to generate high-quality labelled synthetic data, directly addressing the scarcity of real-world fault datasets. The digital twins simulated seven operational states—normal operation, three anomalies (soiling, shading, cracks), and three electrical faults (ground fault, arc fault, bypass diode fault)—capturing essential electrical signatures with sufficient fidelity for model training. Comparative evaluation of classifiers showed that ensemble methods, particularly XGBoost, significantly outperformed simpler models, highlighting its ability to model complex, non-linear relationships from synthetic features. Classification accuracy was high for electrical faults, while anomalies exhibited more inter-class overlap due to their gradual, similar effects on performance.

A key contribution was the probabilistic inference framework, which provided a continuous representation of uncertainty beyond

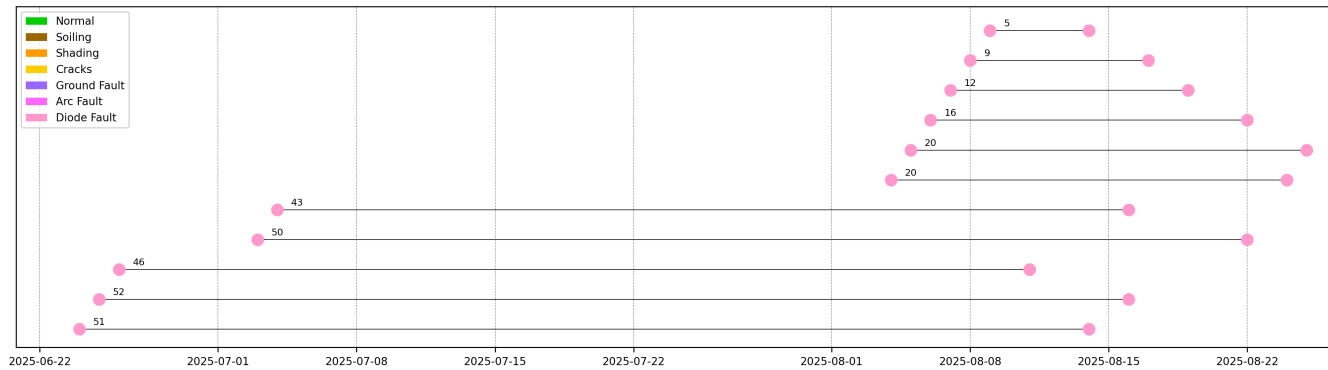


Fig. 11. Cleveland-style Timeline for a Real Time Series in Central Portugal (May–Aug 2025).

binary classification, supporting more nuanced maintenance decisions. When applied to real operational data, the model trained solely on synthetic data transferred effectively, producing class distributions consistent with typical PV system behaviour. For prediction, daily classification probabilities were converted into forward-looking forecasts using linear regression over rolling windows—a computationally efficient approach. The framework captured the temporal progression of degradative conditions, and sensitivity analysis demonstrated how parameter selection could balance prediction lead time and reliability. Overall, the results confirmed that a digital twin-driven ML approach is a practical, scalable solution for enhancing PV system reliability despite limited real fault data.

5.2 Limitations

The proposed framework, while innovative, exhibited several practical limitations affecting its effectiveness and generalisability. Its primary constraint was the reliance on synthetic data for training, which, despite careful parameterisation, simplified real-world stochastic behaviour and introduced a persistent “reality gap.” Validation using real operational data was limited by generic fault labels, preventing definitive quantitative assessment of diagnostic accuracy. Additionally, the 5-minute temporal resolution of both simulated and real data was insufficient to capture high-frequency signatures required for faults such as arc faults.

The framework addressed only seven isolated conditions, excluding other common failures (e.g., potential induced degradation or inverter capacitor ageing) and complex overlapping fault scenarios typical of real installations. Anomaly forecasting relied on linear regression, assuming steady degradation, which failed to reflect the non-linear and abrupt nature of real degradation processes. Moreover, the classifier could not identify unknown fault types, increasing the risk of misclassification.

5.3 Future Work

To transition the framework from a research prototype to an industrial predictive maintenance tool, future work should focus on targeted methodological and practical improvements. The foremost priority is acquiring high-resolution, real-world fault data with

expert-validated labels to enable rigorous model validation. In parallel, the digital twin should be expanded to cover a broader range of failure modes, including compound faults, and its temporal resolution increased to capture high-frequency electrical phenomena. Prognostics should move beyond linear regression to non-linear time-series models, such as LSTMs or transformers, to better represent complex degradation dynamics.

Robustness can be improved by incorporating anomaly detection or open-set recognition to handle unknown fault types. Research should also address deployment constraints through model optimisation techniques—such as pruning and quantisation—for real-time edge inference. Finally, integrating additional data sources, including electroluminescence imaging and advanced weather inputs, would enable a more comprehensive and adaptive diagnostic and prognostic framework suitable for industrial PV systems.

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